

Risk Analysis Report

1. Purpose and Shared Premise

This report is not a generic project-management risk list. It is a requirement-driven risk analysis for the final ARG-Test submission, centered on FR/NFR compliance, evidence traceability, and defense-time interpretability under the course-provided `Requirement_Specification.md`.

The shared team wording for the final phase is:

Our final project is implemented and validated against the provided Requirement Specification of the AutoTestDesign AI App.

Accordingly, the scope of this report covers:

- the requirement-driven AutoTestDesign AI App itself
- the canonical formal result source and its mirrored evidence
- the independent `final_docs/` package
- `demo_web/`, the demo package, and presentation/defense interpretation risks

At this stage, the main risks are no longer “the core system does not exist.” They are now about whether the finished functionality, boundaries, and evidence are explained accurately, consistently, and defensibly.

2. Risk Scoring Method and Evidence Sources

2.1 Risk Scoring Method

Each risk is evaluated with three 1-5 dimensions:

- **Impact**: how much the risk would damage compliance, defensibility, or grading outcome
- **Likelihood**: how likely the risk is to occur given the current repository and document state
- **Detectability**: how difficult it is for the team to catch and correct the risk before submission

The final priority score is:

$\text{Risk Priority} = \text{Impact} \times \text{Likelihood} \times \text{Detectability}$

Priority bands used in this report:

- **High**: ≥ 60
- **Medium**: 36-59
- **Low**: ≤ 35

2.2 Approved Evidence Sources

This report relies only on final-phase approved evidence sources:

- requirement baseline: Requirement_Specification.md
- canonical result root: .local_runs/formal_qwen_novpn
- repository-traceable mirrors: report_assets/final_latex_report/ and report_assets/final_demo_package/frontend_focus/formal_results_snapshot/
- code and execution evidence: src/, demo_web/, tests/, reference_impl/, and final_docs/execution_evidence/
- final document and submission-process evidence: final_docs/ and report_assets/final_demo_package/

2.3 Unified Final Numbers

Whenever the report, PPT, demo, or defense uses quantitative results, the team must use the following final numbers:

- dataset split: dev = 50, test = 16
- full pipeline: avg checker score = 0.959, avg overall coverage = 0.615, avg test count = 7.312
- baselines: Rule-based 0.753 / 0.147 / 4.125; Plain LLM 0.844 / 0.030 / 7.250; Structured No Checker 0.841 / 0.538 / 6.312
- detailed executable evidence: 15 module tests passed; 32 repo tests passed; 100% statement coverage; 100% branch coverage; 4 / 4 mutants killed

These numbers are grounded in report_assets/final_latex_report/main.tex, final_docs/execution_evidence/, and the formal demo snapshot, and must not be replaced by mock outputs or legacy repository leftovers.

3. Requirement-Specification Coverage View

The table below makes the framing explicit: the risks in this report are tied to FR 1.0 ~ FR 7.0 and NFR 4.1 ~ NFR 4.4, not to arbitrary project commentary.

Requirement	Current implementation or evidence closure	Related risks
FR 1.0 Input / Parsing	Supports requirement files, run-text, and batch-csv; see src/main.py, src/input_loader.py, and extended_feature_smoke_summary.md.	R1, R6
FR 1.1 Requirement Structuring	Structured parsing is implemented through src/parser.py, src/schemas.py, and the structured trace format.	R1, R2

Requirement	Current implementation or evidence closure	Related risks
FR 2.0 Risk Analysis & Prioritization	src/risk.py produces requirement-level risk_assessment plus case-level priority promotion.	R1, R2
FR 3.0 Black-Box Test Design	Covers EP, BVA, Decision Table, and State Transition, with baseline, ablation, and final comparison evidence.	R1, R2
FR 4.0 White-Box Test Modeling	In the final scope, this is realized as state-model extraction plus All States and All Transitions planning, with coupon_discount_engine providing executable white-box evidence.	R3
FR 5.0 Test Oracle Generation	Final tables explicitly include Expected Output; repair, export, and execution evidence all depend on expected-result synthesis.	R3, R6
FR 6.0 Output & Export	Supports Markdown / JSON / CSV export; see src/exporter.py and the formal output directories.	R1, R6, R7
FR 7.0 Test Suite Optimization	Multi-candidate reranking, repair, and priority-aware refinement are implemented.	R1, R2
NFR 4.1 Performance	Mock/local-path performance has been validated, but live runtime remains provider-latency-bound.	R5
NFR 4.2 Usability / UI / Traceability	The primary workflow is still CLI plus artifact-based execution, supplemented by the demo_web/ FastAPI UI shell.	R4, R6, R8
NFR 4.3 Security	.env-based secret injection, manifests without API keys, and artifact secret scanning are in place.	R9
NFR 4.4 Maintainability and Technology	The modular Python repo, architecture figure, README, final docs, and demo package together support this requirement.	R6, R10

4. Priority Overview

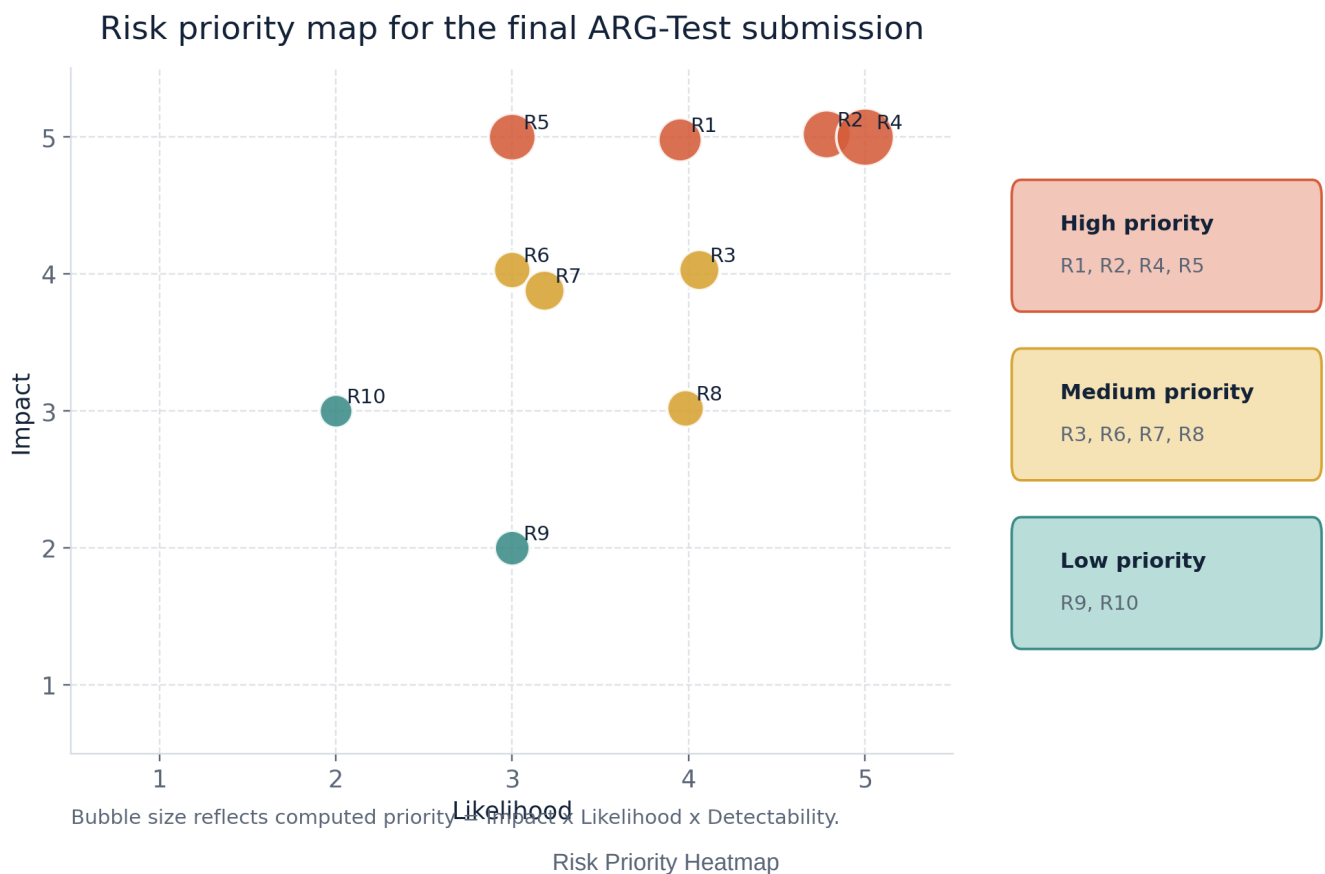


Figure 1 shows the current final-submission posture clearly. The high-priority risks are concentrated around Requirement coverage exposition, claim interpretation, UI boundary, and performance wording, not around complete absence of functionality. That is the expected posture of a technically completed course project entering its final submission phase.

5. Risk Register

Risk ID	Requirement anchor	Risk statement	Impact	Likelihood	Detectability	Priority	Band	Current status	Main mitigation and evidence
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R1	FR 1.0 ~ FR 7.0	The repository already covers input, parsing, risk scoring, black-box design, state modeling, export, and optimization, but if the documents do not explicitly map these capabilities to the Requirement Specification, reviewers may misread the project as “just generating a few test cases.”	5	4	3	60	High	Controlled	Use explicit FR-coverage mapping in the risk report and test plan; rely on src/main.py, src/risk.py, src/state_model.py, src/exporter.py, final_docs/10_feature_closure_status_cn.md, and extended_feature_smoke_summary.md.
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R2	FR 2.0, FR 3.0, FR 7.0	If checker_score, overall_coverage, baseline meaning, and gold-spec usage are not explained precisely, reviewers may misinterpret them as “score equals correctness,” “the model was trained on the dataset,” or “the rule-based baseline comes from an external paper system.”	5	5	4	100	High	Controlled	Always report 0.959 / 0.615 / 7.312 together with the baselines; explicitly state that checker_score != correctness and that the gold spec is an evaluation rubric, not training data; see report_assets/final_latex_report/main.tex and final_demo_script_cn.md.
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| R3 | FR 4.0, FR 5.0 | The title of FR 4.0 can be misread as requiring full source-code-level white-box automation, and coupon_discount_engine may be misread as the main system itself rather than an executable usefulness anchor. | 4 | 4 | 3 | 48 | Medium | Residual but defensible | In the detailed document, explicitly separate “AutoTestDesign app feature under validation” from “selected executable validation module”; see src/state_model.py, final_docs/detailed_test_design_execution/04_detailed_test_design_execution_cn.md, and coupon_discount_engine_execution_summary.md. |

| R4 | NFR 4.2.1 | The specification requires a clean and intuitive UI. If we pretend the project already has a polished standalone web application, we overclaim. If we do not explain the UI at all, reviewers may ask “where is the app?” | 5 | 5 | 5 | 125 | High | Controlled with boundary | The correct wording is that the main workflow remains a CLI-centered prototype, while the final deliverable adds the demo_web/ lightweight FastAPI UI shell for direct text, CSV, state-model, and formal-dashboard demonstration; see demo_web/app.py, demo_web/README.md, and tests/test_demo_web_api.py. |

| R5 | NFR 4.1.1, NFR 4.1.2 | If we directly claim full compliance with “100 requirements within 5 seconds” or “full test generation within 2 seconds,” the statement becomes misleading because live latency depends on the provider. | 5 | 3 | 5 | 75 | High | Residual and accepted | Split the performance claim into two layers: mock/local processing is fast, while live latency remains provider-bound; see final_docs/09_nfr_validation_report_cn.md and nfr_validation_summary.md, which record a local sample average of 0.005 s / requirement. |

| R6 | NFR 4.2.2, NFR 4.4.3 | If requirement IDs, techniques, risk levels, architecture, usage guidance, and evidence paths are not kept in stable alignment, the package may look “rich in materials but poor in traceability.” | 4 | 3 | 3 | 36 | Medium | Controlled | Keep wording and paths synchronized across README.md, final_docs/README.md, 05_evidence_and_submission_checklist_cn.md, and report_assets/final_demo_package/; manifests and structured exports preserve traceability on the output side. |

| R7 | formal result source policy | If the team mixes legacy outputs/reports/test leftovers, mock screenshots, or exploratory runs, the authority of the final numbers will be weakened. | 4 | 3 | 4 | 48 | Medium | Controlled | Anchor the official wording to .local_runs/formal_qwen_novpn; use report_assets/final_demo_package/frontend_focus/formal_results_snapshot/ and report_assets/final_latex_report/ as repository mirrors; see upgrade_pipeline_verification_summary.md and demo_web/app.py. |

| R8 | demo / presentation interpretation | The demo uses mock for stable recording. If the explanation is weak, reviewers may misread it as the final live benchmark or as proof of full provider determinism. | 3 | 4 | 3 | 36 | Medium | Controlled | The demo script explicitly requires the wording “mock is for stable interaction, frozen formal result represents final quality”; see report_assets/final_demo_package/demo_handoff_cn.md and final_demo_script_cn.md. |

| R9 | NFR 4.3.1 | If .env, token screenshots, or terminal history with secrets enter the submission package, that creates a low-frequency but embarrassing security issue. | 2 | 3 | 4 | 24 | Low | Controlled | Secrets are injected through environment variables; manifests do not record API keys; artifact and report directories have been secret-scanned; see 09_nfr_validation_report_cn.md and nfr_validation_summary.md. |

| R10 | NFR 4.4.1, NFR 4.4.2, NFR 4.4.3 | If the project does not present its modern stack, modular architecture, automated tests, and documentation assets as one coherent chain, its maintainability and technology choices may be undervalued. | 3 | 2 | 5 | 30 | Low | Controlled | Use the layered structure of src/ / experiments/ / tests/ / final_docs/ / demo_web/, the 32 passed regression state, the architecture figure, the README, and the demo package together to support NFR 4.4. |

6. High-Priority Risk Analysis

6.1 R1: FR Coverage Exposition Risk

This is not a “missing feature” problem. It is a “finished features may still look unfinished if they are not explained against the requirement map” problem. The repository already supports:

- FR 1.0: run, run-text, batch-csv
- FR 2.0: risk scoring and priority promotion
- FR 3.0: EP / BVA / Decision Table / State Transition
- FR 4.0: state-model extraction, All States, and All Transitions
- FR 5.0: Expected Output synthesis
- FR 6.0: Markdown / JSON / CSV export
- FR 7.0: reranking, repair, and prioritization

The mitigation is therefore not to rebuild functionality, but to keep every final document aligned with the FR structure.

6.2 R2: Metric and Comparison Interpretation Risk

This is one of the most likely defense-time risks. The safest explanation is:

- checker_score measures contract consistency, not final correctness
- overall_coverage measures gold-spec obligation coverage and complements checker_score
- the rule-based baseline is our own deterministic heuristic non-AI baseline
- the gold spec is a manually derived evaluation rubric, not training data

If these statements stay consistent across the risk report, PPT, and demo script, R2 remains a controllable interpretation risk rather than a technical flaw.

6.3 R4: UI Boundary Risk

NFR 4.2.1 is one of the most sensitive non-functional items in this final project. The most accurate wording is:

- the primary workflow is still CLI + artifact-based workflow
- the final defense package adds the demo_web/ lightweight web UI backed by FastAPI
- it already supports direct text input, CSV import, a formal-result dashboard, and state-model generation
- it is a prototype-grade interface, not a large polished product frontend

This wording avoids both extremes: denying the UI entirely and overclaiming a mature web product.

6.4 R5: Performance Claim Risk

The performance section must not be written as if the project already satisfies an industrial live end-to-end SLA. The defensible conclusion is:

- the local/mock path is fast enough for course-project validation and demonstration
- actual live runtime is mainly constrained by provider latency
- the final evaluation emphasizes functionality, coverage, usefulness, and evidence-backed validation

This is more honest and more technically correct than simply repeating the hard numbers from the specification without qualification.

7. Medium-Priority Risks and Accepted Boundaries

7.1 R3: Interpreting FR 4.0 and the Executable Module Correctly

This boundary should not be hidden; it should be explained clearly:

- `state-model extraction` is the direct `AutoTestDesign` app implementation for FR 4.0
- `coupon_discount_engine` is not the whole system
- its role is to show that the generated or curated test design can be translated into concrete black-box and white-box execution evidence

The supporting hard evidence is already strong:

- 15 passed
- 32 passed
- 100% statement coverage
- 100% branch coverage
- 4 / 4 mutants killed

7.2 R6 and R7: Traceability and Canonical Result-Source Risk

These two risks share the same root problem: the project has many materials, but it must maintain only one official final wording. The submission should therefore follow this discipline:

- the requirement source is always `Requirement_Specification.md`
- official numbers are always anchored to `.local_runs/formal_qwen_novpn`
- repository figures, the demo dashboard, and the LaTeX report are mirrors or re-presentations of that official source
- `legacy_outputs/reports/test` snapshots must not be cited as final evidence

7.3 R8: Demo and Defense Misinterpretation Risk

The demo only needs to prove three things:

- the tool truly runs
- the outputs are truly structured

- the final project truly has formal results and executable evidence

It is not a live benchmark, and it is not a provider-determinism demonstration. If that explanation remains intact, the risk stays manageable.

8. Honest Boundaries That Should Remain Visible

The following points should remain visible in the final submission rather than being polished away:

- the current UI is a lightweight web UI plus the original CLI workflow, not a polished standalone product frontend
- current performance evidence mainly proves that the local/mock path is fast enough; live latency remains provider-bound
- the project implements a seed-controlled pipeline, repeatability studies, and replay, but it must not claim that the upstream live provider is fully deterministic
- `checker_score` is not correctness itself, and the `gold_spec` is not training data
- `coupon_discount_engine` is an executable usefulness anchor, not a substitute for the whole AutoTestDesign app

Keeping these boundaries explicit strengthens the credibility of the final submission.

9. Conclusion

From a final-submission perspective, the current risk posture of ARG-Test is positive. The core FR items and main NFR items are already supported by implementation and evidence, and the main remaining challenge is not “how to build a new system,” but “how to avoid being misunderstood.”

The highest-priority risks are concentrated in FR coverage exposition, claim interpretation, UI boundary, and performance wording, and each of these already has a clear mitigation path with repository-backed evidence.

Therefore, the conclusion of this report is not that the project is still highly risky. It is that ARG-Test has already reached a defensible final state, and that the remaining risks are controllable as long as the team keeps the same requirement-driven explanation framework across the documents, demo, PPT, and defense.